HOW TO FOLLOW CONNECTIONS: each pin stub ends at a net label; identical labels are electrically connected. A0–A13: U1–U3; A0–A12 also U20. D0–D7: U1, U2, U3 and U20. Click a label in KiCad to highlight its full net.

CPU/VIA: A0–A15, D0–D7, RWB, IRQ_N. CPU PHI2 is an input; PHI2O drives the VIA clock.

SRAM: A0–A13 direct; U5 selects CPU A14 and fixed zeros or BANK_Q[3:0] for SRAM A14–A17. A18 is grounded, using half of this 512Kx8 candidate.

U7 resets the bank to zero; U6 holds or loads D0–D3 when BANK_REG_WRITE is true. U8 inverts PHI2O and RWB.

U4/U9 decode address pages. Socketed DIP U10/U12 generate RAM select and PHI2-qualified active-low CE/OE/WE; OE and WE are mutually exclusive.

Socketed DIP U15–U16 and U18 generate bank-window select, qualified bank-register writes and PHI2-qualified VIA CS1; VIA CS2_N is tied low.

Text screen: reserve SRAM $7800–$7FFF as a 25x80 row-major character buffer; ROM sends it to the Pico through the VIA. No character ROM or video hardware.

Power: J1 accepts externally regulated +3V3 only; no regulator is fitted. C19 is bulk bypass; C4–C18 are IC bypass capacitors.

Clock: 1.000MHz crystal with socketed 74HCU04 Pierce oscillator, series drive resistor and buffer. Verify crystal startup and CPU clock limits on hardware.

Reset: pull-up, 100nF power-on delay and pushbutton; CPU RDY/NMI/BE/SO and IRQ have defined pull-ups. Validate reset timing from the exact CPU datasheet.

Pico is external: J2 pins 1–8 GP0–GP7, 9 REQ, 10 ACK, 11 DIR, 12 OE_N, 13–14 GND. Pico is separately USB-powered; provide common ground.

U19 SN74HC245N routes Pico A0–A7 to VIA B0–B7; DIR=BUS_DIR, OE_N=BUS_OE_N. Bus direction/output-enable reset defaults are passive.

U20 AT27LV256A: A0–A12 to CPU A0–A12; A13/A14 strapped low for an 8KiB image at $E000–$FFFF. /CE=PAGE_7_N; /OE=PHI2-qualified read.

U20 is shown as a socketed DIP-28 3.3V OTP EPROM legacy candidate; confirm the genuine DIP orderable variant and availability before PCB. Program reset vector at $FFFC/$FFFD.

NOT fabrication-ready: exact EPROM package/sourcing, oscillator and reset timing, power budget/sequencing, Pico harness, BOM and board mechanics need review.

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NOT PCB-ready: verify timing, power, bus interface, BOM and mechanics
Socketed-DIP 3.3 V design draft; legacy OTP EPROM sourcing TBD